

COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

FYP-I PROJECT PROPOSAL

IntelliSafe

Real-Time Working Site Safety Monitoring
using Edge Computer Vision

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Abstract

What we are building

IntelliSafe is an edge-based safety monitoring system for industrial and construction work sites. A camera connected to an **NVIDIA Jetson Nano** runs a single **YOLOv8n** detector that finds workers, their personal protective equipment (helmet and vest), and fire or smoke. A **compliance engine** links each helmet and vest to the worker wearing it, checks the safety rules, and keeps a rolling compliance score. When a violation continues, it sends one alert instead of repeating it on every frame, and it saves an image as evidence.

Why it matters

These events are published over **MQTT** to a backend that stores them, streams them live to a supervisor **web dashboard**, and generates **PDF compliance reports**. Occasional manual inspection is replaced with continuous, evidence-based monitoring. Violations are caught as they happen, and management gets data to act on before an accident occurs.

Introduction

Context

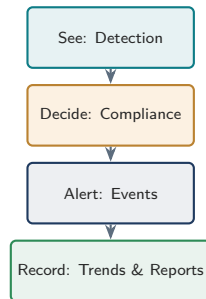
- ▶ ILO estimates $\approx$ 3 million work-related deaths and 395 million non-fatal injuries every year [1].
- ▶ Construction and manufacturing account for a large share; many incidents are linked to missing PPE and to fire or smoke that goes unnoticed.
- ▶ Safety today relies on *human* supervisors doing *periodic* rounds.

Opportunity

- ▶ Low-cost edge AI hardware can now run real-time object detection beside the camera, so video does not have to leave the site [6].
- ▶ Deep-learning PPE detection is proven in research [2, 3]; what sites lack is an integrated, deployable *system* around it.

IntelliSafe closes that gap

The camera detects workers, their helmets and vests, and any fire or smoke. The compliance engine checks whether each worker is wearing the required helmet and vest. The platform alerts the supervisor, stores image evidence, and records the trend for analysis and reporting.



Current practice is ineffective

- ▶ **Coverage gaps** — inspections are periodic, so a missing helmet between rounds goes unnoticed.
- ▶ **Human limits** — one supervisor cannot watch every worker for a whole shift.
- ▶ **No evidence** — when an incident happens, there is rarely a time-stamped image.
- ▶ **No trend data** — management cannot see whether violations are rising or which shift is the weak point.

What is needed

- ▶ Continuous, automated observation of the work site.
- ▶ Immediate detection of missing helmets and vests, and of fire or smoke.
- ▶ Instant alerts to the supervisor.
- ▶ Time-stamped image evidence for every violation.
- ▶ Compliance trends and periodic reports.

Problem statement. *Design and implement a low-cost edge computer-vision platform that monitors a work site in real time, detects PPE non-compliance and fire or smoke, and delivers alerts, image evidence, and compliance analytics to site supervisors.*

- i. **Detection.** Train a YOLOv8n model to detect **workers**, **helmets**, **vests**, **fire**, and **smoke** with good accuracy.
- ii. **Compliance engine.** Link each helmet and vest to the worker wearing it, check that both are worn, keep a compliance score, raise one alert per continuing violation, and save an image as evidence.
- iii. **Edge deployment.** Run the model and the compliance engine in real time on an NVIDIA Jetson Nano.
- iv. **Event delivery.** Send violation and hazard events from the Jetson to the backend over MQTT.
- v. **Backend and storage.** Store events, compliance history, and evidence images in a database.
- vi. **Dashboard and reports.** Show live alerts and compliance trends on a web dashboard, and generate PDF compliance reports.

The project maps to three Sustainable Development Goals.

SDG	Goal	How IntelliSafe contributes
SDG 3	Good Health and Well-being	It spots missing helmets and vests and detects fire or smoke early, so preventable injuries are less likely.
SDG 8	Decent Work and Economic Growth	It supports safe working environments by monitoring PPE use continuously instead of relying on occasional inspections.
SDG 9	Industry, Innovation and Infrastructure	It uses low-cost edge AI to bring real-time, evidence-based safety monitoring to industrial and construction sites.

Software

- i. An annotated dataset and a trained YOLOv8n model for worker, helmet, vest, fire, and smoke
- ii. An edge application on the Jetson Nano that runs detection and the compliance engine and saves image evidence
- iii. An MQTT link that sends events from the Jetson to the backend
- iv. A backend with a database for events, compliance history, and evidence images
- v. A web dashboard for live alerts and compliance trends, plus PDF compliance reports

Documentation & demonstration

- ▶ Project documentation
- ▶ FYP report and final presentation
- ▶ Live end-to-end demonstration on the Jetson Nano with a camera

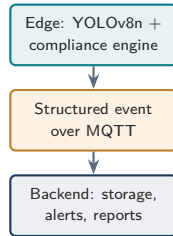
Methodology

Two parts that meet at one fixed handoff

- ▶ **Edge:** a camera connected to a Jetson Nano runs the YOLOv8n detector and the compliance engine.
- ▶ The compliance engine checks each worker for a helmet and vest, raises one alert per continuing violation, and saves an image as evidence.
- ▶ Each result is packed into a structured event and sent over **MQTT** to the backend.
- ▶ **Backend:** stores the events and evidence, pushes live alerts to the supervisor dashboard, and generates PDF compliance reports.

Parallel development

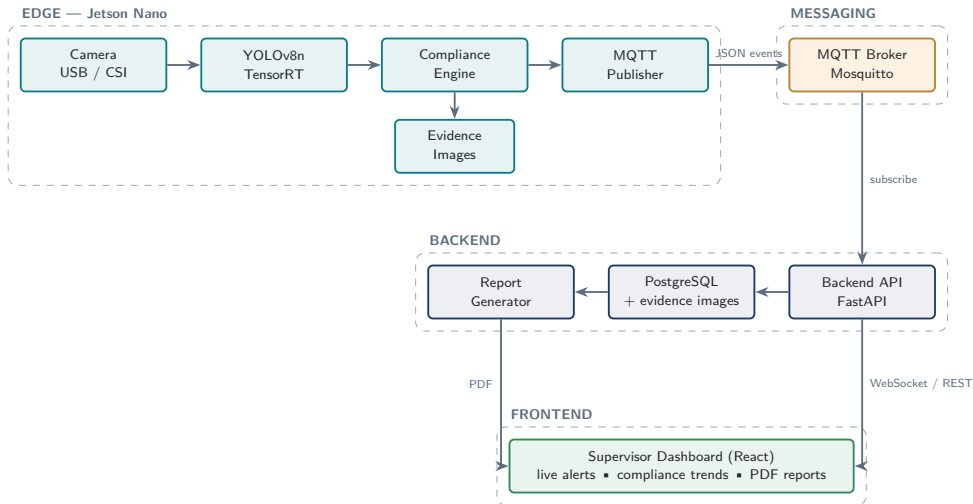
The event format is agreed on at the start, so model training and backend development go ahead in parallel. Until the Jetson is connected, a simple **event simulator** feeds the backend test events.



	Typical PPE-detection projects	IntelliSafe
Scope	Detect helmet / vest in images or video	Detection plus per-worker compliance checks, fire and smoke alerts, evidence, trends, and reports
Deployment	Desktop GPU or cloud	Edge (Jetson Nano) — video stays on site
Output	Bounding boxes on screen	Structured events over MQTT with image evidence
Rules	None	Helmet and vest required for every worker; fire and smoke raise hazard alerts
Alert handling	Frame-by-frame	One alert per continuing violation , with priority levels
Analytics	None	Compliance trends and PDF reports

Contribution. Detection itself is well studied [2, 3]. The contribution of this FYP is a complete, deployable *safety-monitoring system* built around a small edge detector.

Methodology — System Diagram



One model, three detection tracks

Track	Class	Used for
Worker	worker	Count, track, anchor for PPE check
PPE	helmet	Violation: helmet_missing
PPE	vest	Violation: vest_missing
Hazard	fire	Critical hazard event
Hazard	smoke	Critical hazard event

Model & training

- ▶ **YOLOv8n** [4, 5] — the nano variant fits the Jetson Nano
- ▶ Input 640×640 , fine-tuned from COCO weights
- ▶ Export `.pt` → ONNX → TensorRT

Dataset

- ▶ Public PPE datasets (Roboflow / Kaggle), fire and smoke data (D-Fire [7]), and self-collected frames
- ▶ YOLO annotation format; train / validation / test split

1. PPE-worker matching

- ▶ For each worker box, find the helmet and vest boxes that overlap it (IoU)
- ▶ Keep a stable worker_id across frames (simple tracker)

2. Safety rule

- ▶ Helmet and vest are both required for every worker
- ▶ Fire or smoke raises a hazard alert straight away, with no worker check

3. Compliance score

$$\text{Score} = \frac{\sum \text{compliant worker-frames}}{\sum \text{worker-frames}} \times 100$$

over a rolling window of recent frames.

4. One alert per violation

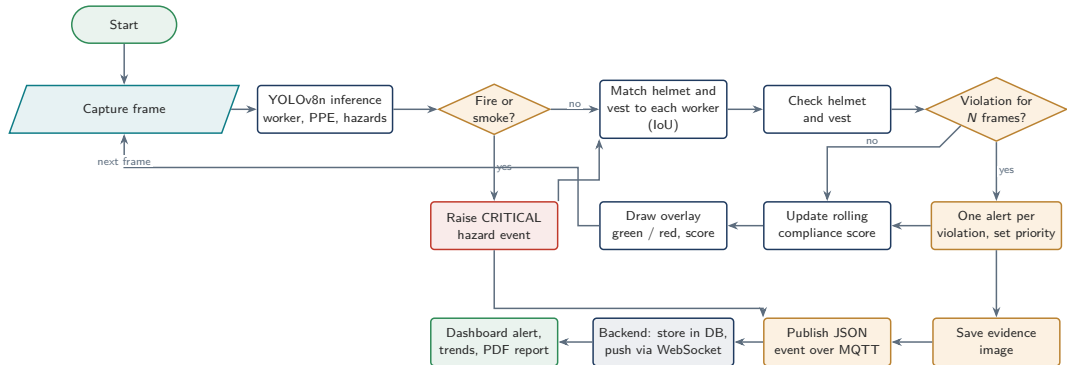
- ▶ A violation must persist for N consecutive frames before an alert is raised; then one alert per worker per violation type, instead of one on every frame

5. Priority & action

Event	Priority	Action
Fire / smoke	CRITICAL	Immediate alert
Helmet and vest missing	HIGH	Dashboard alert
Helmet or vest missing	MEDIUM	Alert + log
Low compliance score	LOW	Flag in report

6. Evidence — on each event, save the annotated frame and a record (time, worker, class, confidence, image path).

Methodology — Flow Chart



Event format (MQTT, JSON)

```
{
  "type": "alert" | "metrics" | "heartbeat",
  "event": "helmet_missing" | "vest_missing" |
    "fire" | "smoke" | "compliant",
  "priority": "CRITICAL" | "HIGH" | "MEDIUM" | "LOW",
  "timestamp": "ISO-8601",
  "worker_id": 7,
  "confidence": 0.92,
  "compliance_score": 78,
  "detections": [{"class", "bbox", "confidence"}],
  "image_path": "evidence/.../frame_001.jpg"
}
```

Topics ppe/alerts, ppe/metrics, and ppe/heartbeat — QoS 1, with buffering on the Jetson if the network drops [8].

Backend (FastAPI + PostgreSQL)

- ▶ Subscribes to the broker; validates and stores events
- ▶ REST: alerts, compliance history, reports
- ▶ WebSocket: pushes live alerts to the dashboard
- ▶ Evidence images stored and linked from the database

Dashboard (React)

- ▶ **Live** — compliance gauge, alert feed with evidence
- ▶ **Trends** — compliance over time, violation breakdown
- ▶ **Reports** — date range → PDF (ReportLab)

Edge

Device	Jetson Nano
Camera	USB / CSI, 720p
OS	JetPack (L4T)
Language	Python
Detection	Ultralytics YOLOv8
Acceleration	CUDA, TensorRT
Vision	OpenCV
Messaging	paho-mqtt

Backend

Broker	Eclipse Mosquitto
API	FastAPI (Python)
Realtime	WebSockets
Database	PostgreSQL
Reports	ReportLab
Testing	pytest

Frontend & tooling

Framework	React + Vite
Charts	Recharts
Styling	Tailwind CSS
Annotation	Roboflow, CVAT
Training	PyTorch, Colab
Version control	Git, GitHub

Metric	Target	How measured
Detection mAP@0.5 (test set)	≥ 0.70	Ultralytics val
Per-class precision / recall	reported	Confusion matrix
Inference speed on Jetson Nano	real time	On-device benchmark
Alert delay, camera → dashboard	measured	Timestamp difference
False alerts per hour	reported	Live test session

Test scenarios

- ▶ Compliant worker → green box, no alert
- ▶ Helmet removed → red box, alert, evidence saved
- ▶ Two workers, one non-compliant → per-worker result
- ▶ Fire / smoke video as camera input → CRITICAL alert on the dashboard
- ▶ Network drop → events buffered and delivered on reconnect

Uzair — Perception

- ▶ Dataset collection and annotation
- ▶ YOLOv8n training, evaluation, and export
- ▶ Compliance engine: helmet and vest matching, score, one alert per violation
- ▶ Evidence capture and the handoff dictionary

Muhammad Ahmad — Platform

- ▶ Jetson Nano setup and model deployment
- ▶ MQTT publisher and broker
- ▶ Backend services and database
- ▶ Web dashboard and PDF reports

Joint

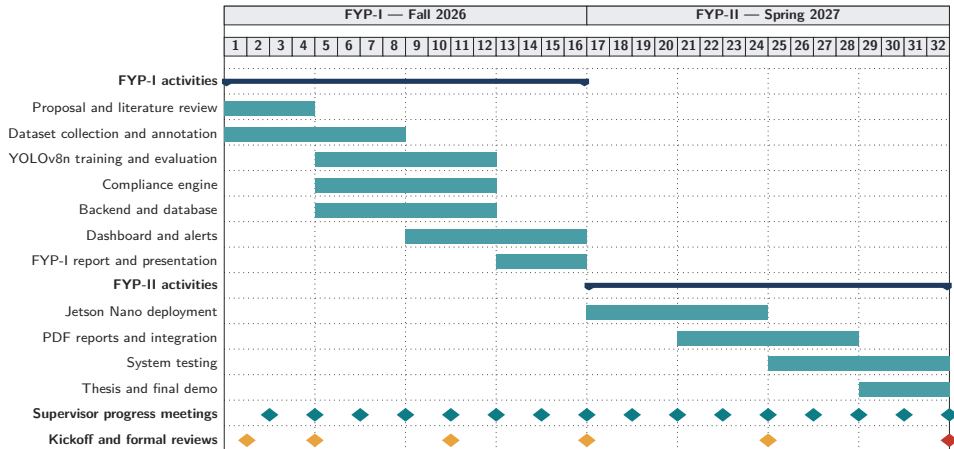
Event format ■ integration ■ testing ■ final demonstration and defence

Risk	Impact	Mitigation
Limited data for fire / smoke	Low accuracy on hazard classes	Combine public datasets (e.g. D-Fire) with self-collected frames; helmet and vest remain the core classes
Jetson Nano too slow	Missed frames, late alerts	TensorRT export, frame skipping; fall back to laptop inference running the same code
Jetson software stack (old JetPack, Python version)	Deployment delays	Start Jetson set-up at the beginning of FYP-II; use NVIDIA PyTorch wheels or ONNX Runtime
Model not ready when the backend is	Blocked integration	Event simulator publishes realistic MQTT events so the backend and dashboard can be built independently
Network failure during demo	Alerts do not reach the dashboard	Local hotspot; MQTT QoS 1 with buffering on the Jetson
Live fire demo is unsafe	Cannot show the CRITICAL path	Recorded fire / smoke video used as camera input

Timeline



Timeline — Gantt Chart (Two Semesters)



◆ progress meeting every two weeks ◆ kickoff (week 1) and formal reviews (weeks 4, 10, 16, 24) ◆ final defence (week 32)

Timeline — Tentative Meeting Schedule with the Supervisor

Week	Meeting	Purpose
1	Kickoff	Agree the scope: worker, helmet, vest, fire, and smoke
4	Proposal review	Proposal approved; dataset sources and tools confirmed
10	Mid-term (FYP-I)	First trained model; compliance engine running on a laptop
16	FYP-I evaluation	Laptop demo from camera to dashboard; FYP-I report
24	Mid-term (FYP-II)	Model running on the Jetson Nano; live events stored
32	Final defence	Full live demo, PDF reports, thesis, and demo video
Every 2 weeks	Progress meeting	Work done, problems, and the plan for the next two weeks

Timeline — Objectives to Achieve during the Year

Period	Weeks	Objectives	Evidence shown to supervisor
FYP-I — Fall 2026			
First half	1–8	Proposal, literature review, dataset collection and annotation, backend set-up	Approved proposal; labelled images
Second half	9–16	Train YOLOv8n; build the compliance engine; basic dashboard	Laptop end-to-end demo; FYP-I report
FYP-II — Spring 2027			
First half	17–24	Deploy on the Jetson Nano; send live events to the backend; store evidence	Live demo from the Jetson to the dashboard
Second half	25–32	Improve the model; PDF reports; system testing; thesis	Final demo; test results; thesis

References

- [1] International Labour Organization, *A Call for Safer and Healthier Working Environments*, Geneva, 2023.
- [2] N. D. Nath, A. H. Behzadan, and S. G. Paal, "Deep learning for site safety: Real-time detection of personal protective equipment," *Automation in Construction*, vol. 112, 103085, 2020.
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- [4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, real-time object detection," in *Proc. IEEE CVPR*, 2016, pp. 779–788.
- [5] G. Jocher, A. Chaurasia, and J. Qiu, *Ultralytics YOLOv8*, 2023. <https://github.com/ultralytics/ultralytics>
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- [7] P. V. A. B. de Venâncio, A. C. Lisboa, and A. V. Barbosa, "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices," *Neural Computing and Applications*, vol. 34, pp. 15349–15368, 2022.
- [8] OASIS, *MQTT Version 5.0*, OASIS Standard, March 2019. <https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html>

Thank You

Questions & Discussion

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